

The story of Agni

India's intercontinental ballistic missile (ICBM) program has reached a new height. We look back on Agni V's journey and gaze into its future

FIFA 13 to be more realistic

A brand new technology called DI3D, adopted by FIFA13, provides highly realistic features to the players inside a game <http://digit.in/J6TZKh>

A clear picture

With image deblurring you could soon be saying goodbye to those blurry shots which have ruined hundreds of your precious memories

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Blurry images are the bane of photographers, be they NASA astronomers trying to capture the secrets of the universe, or an amateur photographer trying to capture an image of his/her delinquent puppy. Not very long ago at Adobe MAX 2011, Adobe demonstrated a seemingly miraculous cure to this problem. You load a blurred image into Photoshop, analyse the image, hit Deblur, and voila! You have a crystal clear image in almost no time at all.

This cure to blurry images isn't as simple as demonstrated though. Adobe itself was quick to point out that its algorithm works in only certain scenarios and the images it demonstrated were analysed to determine the best "deblurring" method long before they were shown to the public (<http://digit.in/JrpXTW>). The technology is still in its infancy and is something that will involve a lot of time and effort to get right. Why is deblurring so difficult to achieve? One would have thought that with the oodles of computing power available, this should have already been in place by now.

How was it done?

Adobe's algorithms are top secret of course, but various deblurring algorithms have existed for decades now and have been used by forensic experts, various space agencies and even military organisations for more than a decade now. In fact, digital image stabilization, found in a lot of current digital cameras, uses a variant of these algorithms.

Let's start with the first step – you load your blurred image into a future release of Photoshop or similar image editing tool. The first thing the program does is analyse the image and try to determine its Point Spread Function (PSF). The PSF is



Before and after images showcased at Adobe MAX 2011

nothing more than the path that's followed by your image sensor when the image was being taken. To put it more simply, your shaky hands followed a certain path while the image was being captured, the data captured by the sensor also followed that path and the algorithm tries to determine that path from the image. The PSF can even be the blurring caused by diffraction, an out-of-focus image, etc.

A simple analogy

Consider that you have a paintbrush that is fixed to a wall. If you were to hold up a paper to that brush, your hands would inevitably shake, resulting in a squiggle on the sheet of paper. You feed this paper into a computer and ask it to analyse the squiggle and figure out what the shape of the brush's tip was. This, while still a bit complicated, might be managed. Now imagine you had a million brushes attached to a wall, some moving at random, others of varying colours and at various angles, and you try to hold another sheet of paper over this. The resulting squiggle would be horrendous to behold (or maybe modern art) and you then ask the computer to figure out what the original pattern was.

It's a monumental task, and despite the smartest algorithms in the world, this task would take ages.

Obviously, each paintbrush represents a point of light from the image you're trying to capture, the paper is the sensor and your hands (or something else) are the image



Before (without the sensor attached)



After (with the sensor attached)